LAB EXPERIMENT RECORD

Set 20: Stock Analysis

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192473018
Date	15 October 2026
Day	Thursday

Dataset: Stock Prices

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

1. Line Chart

Aim

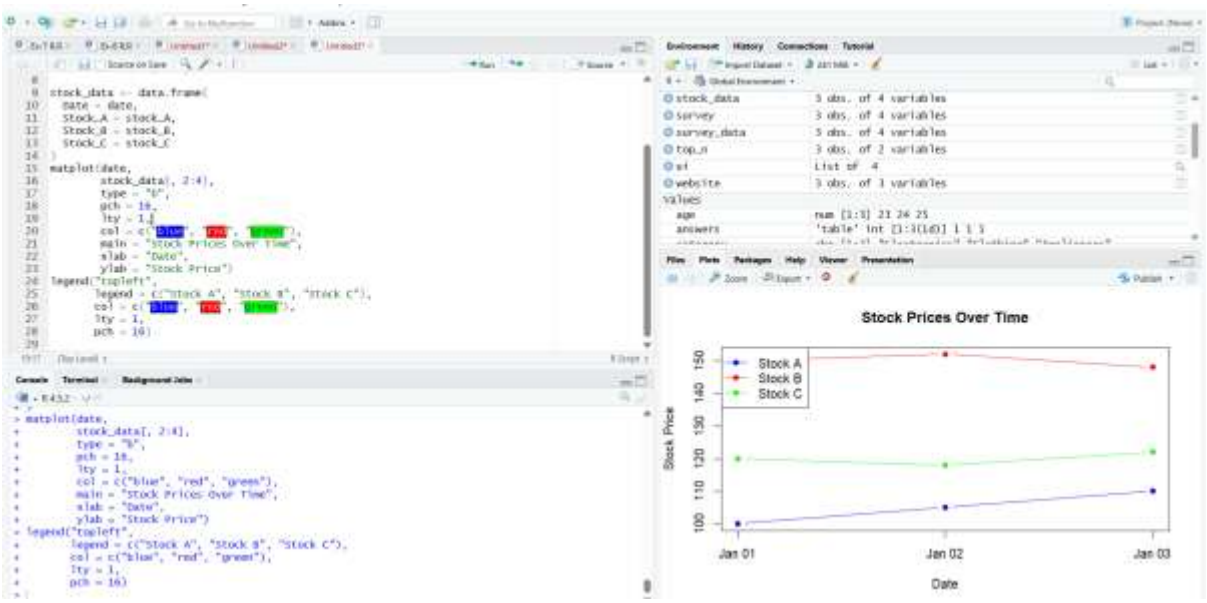
To create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period and label the axes and title the chart.

R Code

```
date <- as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03"))
stock_A <- c(100, 105, 110)
stock_B <- c(150, 152, 148)
stock_C <- c(120, 118, 122)
```

```
stock_data <- data.frame(
  Date = date,
  Stock_A = stock_A,
  Stock_B = stock_B,
  Stock_C = stock_C
)
matplot(date,
        stock_data[, 2:4],
        type = "b",
        pch = 16,
        lty = 1,
        col = c("blue", "red", "green"),
        main = "Stock Prices Over Time",
        xlab = "Date",
        ylab = "Stock Price")
legend("topleft",
       legend = c("Stock A", "Stock B", "Stock C"),
       col = c("blue", "red", "green"),
       lty = 1,
       pch = 16)
```

Output



A line chart showing the stock prices of Stock A, Stock B, and Stock C from January 1, 2023, to January 3, 2023.

Observation

Stock A increased from 100 to 110. Stock B increased initially from 150 to 152 and then decreased to 148. Stock C decreased from 120 to 118 and then increased to 122.

2. Bar Chart Showing Daily Percentage Change in Stock A

Aim

To generate a bar chart showing the daily percentage change in stock prices for Stock A and label the chart elements.

R Code

```
date <- as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03"))

stock_A <- c(100, 105, 110)

percentage_change <- c(
  NA,
  diff(stock_A) / head(stock_A, -1) * 100
)
```

```
bar_values <- percentage_change[-1]

bar_dates <- date[-1]

barplot(bar_values,

        names.arg = format(bar_dates, "%Y-%m-%d"),

        main = "Daily Percentage Change in Stock A",

        xlab = "Date",

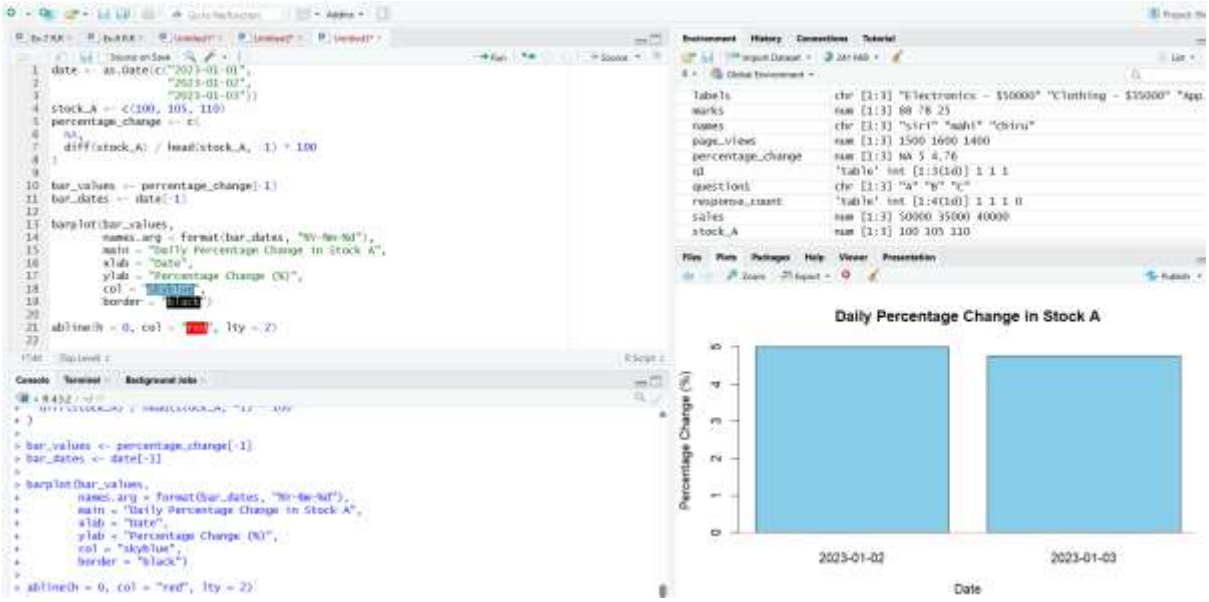
        ylab = "Percentage Change (%)",

        col = "skyblue",

        border = "black")

abline(h = 0, col = "red", lty = 2)
```

Output



A bar chart showing the daily percentage changes in Stock A for January 2 and January 3, 2023.

Observation

Stock A increased from 100 to 105 on January 2, showing a percentage increase of 5%. On January 3, it increased from 105 to 110, showing a percentage increase of approximately 4.76%. Therefore, Stock A showed a positive price change on both days.

3. Stock Price Table

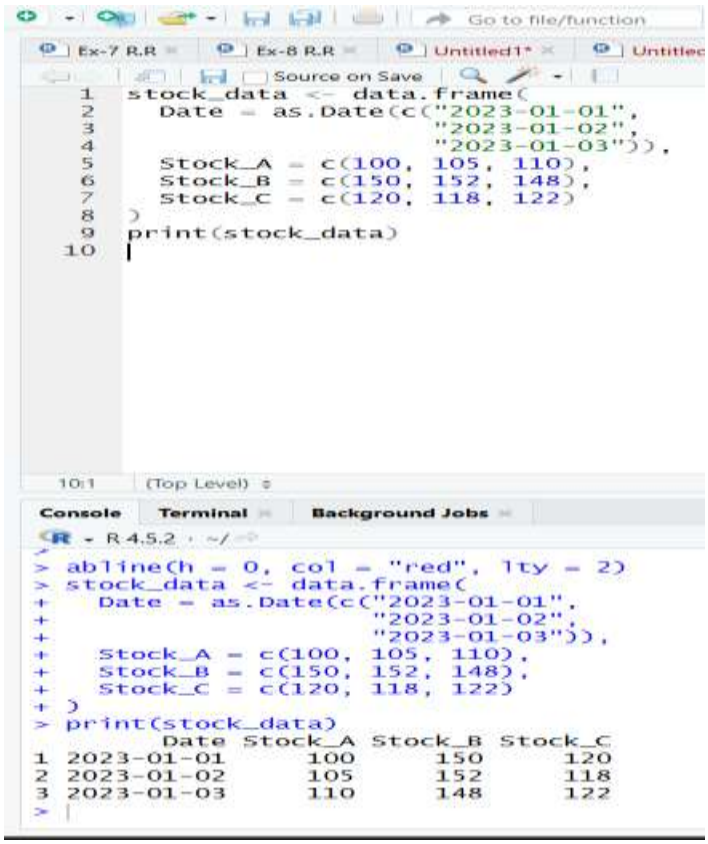
Aim

To create a table displaying the stock price data for each company over the given period and label the table elements.

R Code

```
stock_data <- data.frame(
  Date = as.Date(c("2023-01-01",
                   "2023-01-02",
                   "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)
print(stock_data)
```

Output



Observation

The table displays the stock prices of all three companies for the given dates. Stock B has the highest price on January 1 and January 2, while Stock B and Stock C have prices of 148 and 122, respectively, on January 3, with Stock B remaining the highest.

4. Tableau Dashboard

Aim

To develop an interactive dashboard combining the line chart, bar chart, and stock price table for exploring stock prices.

Tableau Procedure

1. Open Tableau.
2. Import the Stock Prices dataset.
3. Create a line chart using Date and the stock prices of Stock A, Stock B, and Stock C.
4. Add a suitable chart title and label the axes.
5. Create a calculated field for the daily percentage change in Stock A using the formula:

Percentage Change = ((Current Price − Previous Price) / Previous Price) × 100

6. Create a bar chart using Date and the daily percentage change in Stock A.
7. Create a table containing Date, Stock A, Stock B, and Stock C.
8. Select **Dashboard → New Dashboard**.
9. Drag the line chart into the dashboard.
10. Drag the bar chart into the dashboard.
11. Drag the stock price table into the dashboard.
12. Arrange the charts and table neatly.
13. Add suitable titles, labels, and date filters for interactive exploration.
14. Save the dashboard.

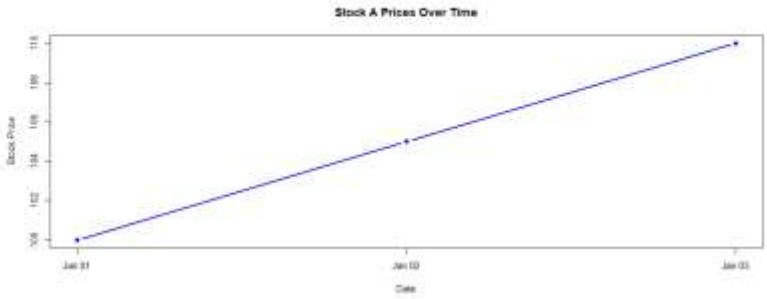
Output

Stock Analysis Dashboard

Select Stock:

Stock A

Stock Price Line Chart



Daily Percentage Change in Stock A



Stock Price Data Table

Date	Stock_A	Stock_B	Stock_C	Percentage_Change
19358.00	100.00	150.00	120.00	NA
19359.00	105.00	152.00	118.00	5.00
19360.00	110.00	148.00	122.00	4.76

An interactive dashboard displaying stock price trends, daily percentage changes in Stock A, and the stock price table for all three companies.